Arista Access Switch Node MAR-ACC-36 MARATHAHALLI CORE Hardware/Environmental Fault Root Cause Identification and Resolution SOP

Node: MAR-ACC-36 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_3641

Description

The failure domain in question pertains to a Hardware/Environmental Fault in the Access role on the MAR-ACC-36 node, situated within the Marathahalli Core site. This failure domain is precipitated by ASIC memory parity errors, line card crashes, and power supply redundancy loss, all of which can be attributed to the intricacies of Arista's EOS (Extensible Operating System) and the underlying hardware architecture. Specifically, the EOS relies on the Arista 7250X-64 switch series, which incorporates a 40GbE switch fabric and 10GbE line cards. The ASIC memory parity errors and line card crashes can be caused by transient hardware faults, software bugs, or environmental factors such as power supply instability or overheating. The loss of power supply redundancy can be caused by a variety of factors, including failed power supply units, power cabling issues, or electrical overloads. Understanding the root causes of these failures is crucial in preventing future occurrences and ensuring the stable operation of the network.

Pre-requisite Checks

- Verify that the Marathahalli Core site is accessible and that remote hands are dispatched to the location

- Check the status of the redundant power supplies and ensure that they are functioning properly

- Verify that the redundant links are operational and that there are no issues with the network connectivity

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to transient hardware faults or software bugs	High
Line card crashes due to power supply instability or overheating	Medium
Loss of power supply redundancy due to failed power supply units or power cabling issues	Low

Diagnostic Commands & Expected Outputs

```
Command: show system internal memory, Expected Output: Memory usage and parity errors information, Description: This command is used to identify any memory parity errors and understand the usage of the system memory.
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```
Command: show system internal linecard, Expected Output: Line card status and error information, Description: This command is used to identify any issues with the line cards, such as crashes or errors.

Command: show system internal power, Expected Output: Power supply status and redundancy information, Description: This command is used to verify the status of the power supplies and ensure that there is redundancy in place.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: no switch 1/1-2, Description: This command is used to remove the faulty ASIC and replace it with a new one. The 'no' keyword is used to prevent any unintended changes to the configuration.

Step 2: Step: Replace the faulty line card, Command: no linecard 1/1-2, Description: This command is used to remove the faulty line card and replace it with a new one. The 'no' keyword is used to prevent any unintended changes to the configuration.

Step 3: Step: Replace the faulty power supply, Command: no power 1/1-2, Description: This command is used to remove the faulty power supply and replace it with a new one. The 'no' keyword is used to prevent any unintended changes to the configuration.

Post-Check Verification

```
Command: show system internal memory, Expected Output: Memory usage and parity errors information, Description: This command is used to verify that the memory parity errors have been resolved and that the system memory usage is within normal limits.

Command: show system internal linecard, Expected Output: Line card status and error information, Description: This command is used to verify that the line card crashes have been resolved and that the line cards are functioning properly.

Command: show system internal power, Expected Output: Power supply status and redundancy information, Description: This command is used to verify that the power supply redundancy has been restored and that the power supplies are functioning properly.
```

Roll-back Procedures

- Step: Re-apply the original configuration, Command: copy running-config startup-config, Description: This command is used to re-apply the original configuration and revert any changes made during the resolution process.

- Step: Revert to the previous configuration snapshot, Command: rollback snapshot previous, Description: This command is used to revert to the previous configuration snapshot and discard any changes made during the resolution process.